

ASSIGNMENT

ROLLING COIN ON A BOUNCING PLATFORM

Part 1: Environment & Platform

Task 1.1 – Ground & Platform

- Add a Plane as the ground.
- Create a platform elevated above the ground.

Task 1.2 – Bouncy Material

- Create a Physic Material
- Apply this material to the platform's collider to make the coin bounce when it hits the platform.
- The platform will impart a strong upward bounce to anything that lands on it

Part 2: The Coin Object

Task 2.1 – Coin Shape & Appearance

- Create a Cylinder (or a flattened cylinder) to look like a coin.
- Change its material to yellow/gold or orange.

Task 2.2 – Rolling & Bouncing Physics

- Add a Rigidbody component to the coin.
- Ensure Use Gravity is checked.
- Add a Physic Material to the coin's collider
- Position the coin above the platform so that when you press Play, it drops, bounces, and rolls off the platform.

Part 3: Lighting & Particle Effect

Task 3.1 – Lighting & Shadows

- Add a Directional Light angled to cast shadows.
- Ensure the coin casts a shadow on the platform and ground.

Task 3.2 – Coin Sparkle Effect

- Add a Particle System as a child of the coin to make the coin look magical or shiny.
- Configure the particles to be small, yellow dots or sparks that appear continuously.
- The effect should move with the coin as it rolls.

Part 4: Bounce & Roll Demonstration

Task 4 – Automatic Physics Demo

- The coin must:
 - Fall onto the platform and bounce at least twice (visible vertical movement).
 - Roll off the edge of the platform and fall to the ground.
- Adjust the coin's starting position and rotation so that gravity and bounciness naturally cause it to roll sideways off the platform.